



Highly efficient and robust Cu catalyst for non-oxidative dehydrogenation of ethanol to acetaldehyde and hydrogen

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ABSTRACT

While acetaldehyde is industrially produced in the Wacker process by homogeneous catalysts, a heterogeneous catalyst is so far elusive due to rapid deactivation stemming from metal sintering and coke formation. This work reports on a highly efficient, robust and stable Cu/SiO₂ catalyst for the non-oxidative dehydrogenation of ethanol prepared by the ammonia evaporation method. The highly dispersed Cu-particles are distinguished by outstanding thermal stability with very little sintering up to 600 °C. This was attributed to the strong interactions between the copper crystallites and SiO₂ arising from the immediately formed copper phyllosilicate phase. Deactivation due to coke formation was successfully addressed by optimizing the metal loading. The CH_x/Cu ratio in the used catalyst decreased by a factor of 10 as the copper loading was increased from 0.5 to 10 wt%. Samples with Cu ≥ 10 wt% demonstrated excellent activity of > 90% ethanol conversion and > 98% acetaldehyde selectivity with no observable deactivation for up to 500 h on-stream at 280 °C and a GHSV of 8895 h⁻¹ (34,930 ml g⁻¹h⁻¹).

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1. Introduction

Ethanol is arguably the most promising biomass-derived chemical to supplant traditional fossil feedstocks [1–3]. Because of the incentive-driven and government-mandated increase in bio-ethanol output, it is now becoming economical within the circular economy to use ethanol as base for value chains to chemicals such as acetaldehyde [4–6], acetic acid [7], ethyl acetate [8] and butadiene [2]. Acetaldehyde is an important feedstock for the production of many other high-volume chemicals. In 2016, the annual production of acetaldehyde was valued at US\$1.26 billion [9]. Even though one of the main outlets of acetaldehyde, the oxidation to acetic acid, is diminishing in importance due to the more efficient carbonylation of methanol, other segments of the acetaldehyde market are growing. Most of the growth will be in applications where acetaldehyde is used as a versatile synthon for transformation into pyridines, pentaerythritol, or polymers used by the pulp and paper, pharmaceutical and water treatment industries. Such consumer-near applications will particularly benefit from the availability of bio-acetaldehyde.

Today, most acetaldehyde is produced by the Wacker process where ethylene - obtained from steam-cracking of fossil fuels - is

oxidized by air or O₂ in the liquid phase using an aqueous homogeneous PdCl₂/CuCl₂ catalyst system [10]. A major drawback of this process is the corrosive nature of the chlorine-containing strongly acidic reaction medium which requires exotic materials for the reactor and auxiliary process equipment. Gas-phase processes starting from ethanol with a heterogeneous catalyst are inherently simpler. However, such processes have so far been unsuccessful due to rapid catalyst deactivation, spurring continuing research in this area. A number of heterogeneous catalysts have been researched including carbon [11–14], Al₂O₃ [15], vanadium [16–23], copper [4,24–26], silver [27–29], molybdenum [30], cobalt [6,31] and alloys of gold-copper [32–40]. Hensen's group [35,41–44] found that Au supported on MgCuCr₂O₄ was highly active with a space-time-yield of 583 g_{aldehyde} g_{Au}⁻¹ h⁻¹ at 250 °C. The good activity was attributed to a high surface Cu⁺ fraction and Au⁰-Cu⁺ synergy.

Two gas-phase processes for the conversion of ethanol into acetaldehyde have been described in the literature: (i) the selective oxidation with air and (ii) dehydrogenation. Compared to the oxidative route, dehydrogenation offers distinct advantages, e.g., over-oxidation to acetic acid and CO₂ is avoided, the produced acetaldehyde is easily separated from the byproducts of the reaction and equimolar amounts of hydrogen are obtained as a valuable co-product [45,46]. Furthermore, the non-oxidative dehydrogenation of ethanol is inherently safe whereas mixtures

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of ethanol and O_2 pose serious safety concerns for large-scale industrial process [29]. The mechanism for dehydrogenation of ethanol to acetaldehyde at the surface of the heterogeneous catalyst has been proposed to involve the following steps: (1) dissociative adsorption of ethanol to form ethoxide and proton, (2) cleavage of the α -C-H bond and proton abstraction and (3) hydrogen recombination and desorption of the aldehyde to regenerate the active sites. Step (2) is proposed to be rate-limiting [2,4,15,29,39,47–51], especially for SiO_2 -based catalysts where the high surface area of SiO_2 offers abundant sites for ethanol adsorption, facilitating the formation of the ethoxide species [29,50].

Copper-based catalysts show high activity and selectivity for the non-oxidative dehydrogenation of ethanol to acetaldehyde [24,26,52–57]. Chang et al. [26] reported 100% selectivity to acetaldehyde at 80% conversion using Cu/RHA (rice husk ash) as catalyst at 275 °C. Bueno's group [24,25,58] found that the catalytic activity of Cu was related to the specific electron density which depended on the Cu particle size and the interface between Cu and support. At 225 °C, acetaldehyde was the main product over 20% Cu/ SiO_2 with 41% conversion and 87% selectivity. However, with ZrO_2 as support, the main product was ethyl acetate with 76.2% selectivity.

Unfortunately, the initially high activity of the Cu catalysts deteriorates rapidly within less than 100 h on-stream even at temperatures below 300 °C (Table 1). The deactivation has been attributed to sintering of the Cu particles as well as carbon deposition [26,56,58,60]. Rao et al. observed that Cu crystallites started to aggregate above 280 °C [61,62]. Thus, the main mode of deactivation in Cu-based catalysts appears to be metal sintering rather than carbon deposition. Sintering can be prevented if one succeeds to strongly bind the metal to the support. Therefore, the interaction between the metal particles and the support is essential. In this work, an ammonia evaporation method was used to deposit copper onto the support. This method was first reported by Miyazaki et al. and involves the addition of ammonia to complex the metal ions [63,64]. Subsequent heating evaporates the ammonia accompanied by hydrolysis of the metal-ammonia complex. As stable supported metal catalysts for vapor-phase hydrogenation reactions have been reported [65–68], it seems to be a worthwhile method to explore for the supported Cu catalysts in this study.

2. Experimental

2.1. Catalyst preparation

The catalyst preparation by ammonia evaporation was carried out as follows. For a 10 wt% Cu loading on SiO_2 (PQ Corporation, CS2040), 20 ml of 28% aqueous ammonia was added dropwise into a solution containing 0.366 g $Cu(NO_3)_2 \cdot 2.5H_2O$ in 100 ml deionized water. After stirring for 5 min, 0.90 g SiO_2 was added and the sus-

pension was stirred for 2 h at room temperature followed by another 3 h at 50 °C to evaporate most of the ammonia. After vacuum filtration, the filter cake was washed, dried and calcined in air for 12 h at 600 °C. Samples with 0.5–50 wt% Cu were similarly synthesized. These samples are denoted as x wt% Cu/ SiO_2 -AE where AE stands for ammonia evaporation.

In a modification, a 10 wt% Cu/ SiO_2 sample was synthesized by a combination of deposition precipitation followed by ammonia evaporation (10 wt% Cu/ SiO_2 -DP-AE). Solution A containing 0.366 g $Cu(NO_3)_2 \cdot 2.5H_2O$ dissolved in 50 ml deionized water and solution B with 0.2329 g Na_2CO_3 in 50 ml deionized water were prepared. To solution B kept in a water bath at 60 °C were added solution A and 0.90 g solid SiO_2 . After stirring the suspension of $CuCO_3/SiO_2$ for 30 min, 3.02 ml of 4.25 M aqueous ammonia was added. Stirring was carried out at 0 °C for 30 min, then at room temperature for 1 h and at 50 °C for 3 h to slowly evaporate the ammonia. The solids were recovered by vacuum filtration, dried and calcined as above.

The ammonia evaporation synthesis method was compared with wet impregnation (WI) and homogeneous deposition-precipitation (DP) for 10% Cu/ SiO_2 (Supplementary data). In addition, the role of the support was studied by loading copper onto La_2O_3 (Degussa), MgO (Merck) and ZrO_2 (synthesized) and mesoporous silica-alumina MSU-9 (Si/Al = 9, synthesized) [69,70] by the wet impregnation method. These samples are referenced as 10 %Cu/support-WI.

2.2. Catalyst characterization

X-ray diffraction (XRD) spectra were measured on a Bruker D8 Advance x-ray diffractometer equipped with Cu anode and a Lynxeye XE detector. The 2θ range from 20 to 80° was recorded with 0.02° step size and 1 s dwell time. A dome-type air-tight sample holder was used for samples after H_2 pre-treatment and after reaction. IR spectra were recorded at a resolution of 4 cm^{-1} using a Bruker ALPHA spectrometer equipped with a diamond Attenuated Total Reflection (ATR) unit. Nitrogen adsorption/desorption isotherms were measured using a Micromeritics Tristar 3000 porosimeter. The surface area and pore size distribution were calculated according to the Brunauer-Emmett-Teller (BET) and Barrett-Joyner-Halenda (BJH) model, respectively. Prior to the measurement, samples were degassed at 300 °C (or 120 °C for used catalysts) under nitrogen flow for 5 h. Transmission electron microscopy with energy dispersive X-ray spectroscopy (TEM/EDS) was performed on a JEOL 3100 transmission electron microscope.

Hydrogen temperature-programmed reduction (H_2 -TPR) was measured using a Micromeritics Auto Chem II 2920. After pretreatment at 300 °C for 30 min in He, the sample was heated to 900 °C in 10 vol% H_2 flow and the evolved gases was monitored by a TCD detector. The basicity of the catalysts was investigated by temperature programmed desorption (TPD) of CO_2 . The sample was prere-

Table 1
Catalysts for ethanol dehydrogenation and causes of deactivation.

	Catalyst	Temp. (°C)	Conv. (%)	Sel. (%)	TOS (h)	Deactivation	Ref
1	Cu-Ni alloy	250	26	100	43	Sintering	[5]
2	Cu/ ZrO_2	300	56.9	54.1	12	Sintering	[24]
3	Cu/Rice husk ash (SiO_2)	275	77	100	4	Sintering	[26]
4	Cu/Rice husk ash (SiO_2)	300	81	100	3	Sintering	[59]
5	$CuO-Cr_2O_3/Al_2O_3$	300	50	55	15	Sintering	[52]
6	Cu/ $MgAlO_x$ (hydrotalcite)	300	75	99	18	Coke deposition	[54]
7	KCu/SiO_2	300	60	100	4	Sintering	[56]
8	Cu/ ZrO_2	275	80.7	15	25	Sintering	[58]
9	Cu-Cr-Co/asbestos	275	49.6		100	Coke deposition	[57]

TOS: Time-on-stream.

duced in 10 vol% H₂ at 300 °C for 1 h, then cooled to room temperature in He before exposure to a flow of CO₂ for 15 min. After flushing the sample for 1 h to remove any physically adsorbed CO₂, the temperature was raised to 500 °C at 10 °C/min. The evolved CO₂ was monitored using a quadrupole mass spectrometer (Balzers QMS 200). Thermogravimetric analysis (TGA) was performed on a TA Instruments Discovery. Around 10 mg of the sample was kept at 100 °C for 30 min under a flow of purified air before heating to 800 °C at 5 °C/min. The weight loss due to combustion of deposited carbon on used Cu catalysts and the evolved H₂O and CO₂ were quantified on a Mettler-Toledo TGA/DSC 1 STAR^e system coupled to a mass spectrometer (MS-Pfeiffer QMS 100). The Cu content in the samples was measured by inductively coupled plasma atomic emission spectroscopy (ICP-AES) using an Optima 5300 DV ICP-OES system. The Cu dispersion was determined by N₂O pulse titration at 90 °C in the TPD system. The samples were reduced in 10 vol% H₂ at 300 °C for 1 h before measurements. X-ray photoelectron spectroscopy (XPS) was performed using a VG Scientific ESCA-lab Mark 2 spectrometer with magnesium anode (Mg K_α = 1253.6 eV, 300 W), Omicron EA125 U7 hemispherical electron analyzer and a 7-channel detector. The C 1s adventitious carbon peak at 285 eV was used for charge referencing.

2.3. Catalytic activity

A fixed-bed gas phase continuous flow reaction system was used. Typically, 0.1 g Cu catalyst and 1.5 g glass beads (180 μm, Sigma-Aldrich) were placed in a tubular quartz reactor fitted with a fritted disk (i.d. 10 mm). The catalyst was pre-reduced in a flow of 10 vol% H₂/He at 300 °C for 1 h before feeding ethanol at a rate of 200 μl/h with a syringe pump (New Era, NE-1000 syringe pump) into a carrier gas flow of 30 ml/min He. The temperature was varied from 150 to 300 °C. The effluent was detected by an online gas chromatograph (HP 5890 series II) equipped with TCD and FID detectors. The products were verified by GC-MS. Gas hourly space velocity (GHSV) and contact time of the reactant (*t_c*) were calculated as follows:

$$\text{Gas hourly space velocity, GHSV (h}^{-1}\text{)} = \frac{V_{\text{He}} + V_{\text{EtOH}}}{V_{\text{catalyst}}}$$

where *v* is the flow rate of the gas at the reactor temperature (in ml h⁻¹) and *V_{catalyst}* is the catalyst volume (ml):

$$\text{contact time, } t_c \text{ (s)} = \frac{3600}{\text{GHSV (h}^{-1}\text{)}}$$

3. Results and discussion

3.1. Characterization of supported copper catalysts

The surface area of the 600 °C-calcined Cu/SiO₂ samples prepared by ammonia evaporation was very high (Table 2). Even after loading with 50 wt% Cu, the surface area was 232 m²/g. For the 10 wt% Cu/SiO₂, lower surface areas were obtained for preparations by wet impregnation or deposition precipitation. Based on the ICP results, the loading was very close to the expected Cu weight percentage, showing that the metal was quantitatively deposited onto the SiO₂ support. This was further supported by the colorless supernatant obtained after removal of the loaded sample (Fig. S1).

The SiO₂ support was amorphous. However, XRD patterns of Cu/SiO₂ before calcination contained broad peaks at 30.8, 35, 57.5, and 62.4° which can be assigned to copper phyllosilicate, Cu₂Si₂O₅(-OH)₂ (PDF 00-027-0188) (Fig. S2). Copper phyllosilicate comprises layers of SiO₄ tetrahedra sandwiched between discontinuous layers of CuO₆ octahedra [71]. The formation of this phase is sup-

ported by FTIR measurements [72]. The spectra of the uncalcined samples show a band at 670 cm⁻¹ and a shoulder at 1032 cm⁻¹ which are distinctive of δ_{OH} and ν_{SiO}, respectively, in copper phyllosilicate (Fig. S3). Although the 1032 cm⁻¹ band overlaps with the ν_{SiO} asymmetric stretching band of SiO₂ at 1068 cm⁻¹, it can still be clearly seen that its intensity as well as that of the 670 cm⁻¹ band increase with copper loading, implying the formation of increasing amounts of copper phyllosilicate. Calcination results in the decomposition of copper phyllosilicate to CuO and SiO₂ [71–73]. After calcination at 400 °C, both bands could still be observed but after heating to 600 °C, the δ_{OH} band was no longer observable due to removal of the OH groups. No bands due to copper phyllosilicate could be observed after calcination at 800 °C.

The XRD patterns of 600 °C-calcined Cu/SiO₂-AE samples with loadings up to 30 wt% Cu were amorphous (Fig. S4). The absence of any diffraction peaks of CuO or any Cu-related phase suggests that the metal was highly dispersed on the support. Only at the highest loading of 50 wt% Cu were the diffraction peaks of CuO detected. In contrast, the XRD patterns of 10 wt% Cu/SiO₂ prepared by wet impregnation (WI) and deposition-precipitation (DP) clearly showed diffraction peaks of CuO (PDF 00-048-1548) (Fig. 1). Using the Scherrer equation, the average CuO crystallite size of the DP- and WI-samples was estimated to be 22.8 and 83.5 nm, respectively. Interestingly, no CuO phase could be detected in 10 wt% Cu/SiO₂-DP-AE where the sample was treated by ammonia evaporation after deposition precipitation. The TGA profiles also reflect the differences in the nature of the copper precursors arising from the different synthesis methods (Fig. S5). For 10 wt% Cu/SiO₂-WI, loss of water occurred below 120 °C followed by a sharp weight loss at about 240 °C due to the decomposition of bulk Cu₂(OH)₃NO₃ [74–76]. In comparison, the TGA profile of 10 wt% Cu/SiO₂-AE showed a continuous weight loss till 650 °C.

In the HRTEM measurements for 10 wt% Cu/SiO₂-AE calcined at 600 °C, no CuO particles could be observed even at 5 nm resolution, supporting the high metal dispersion (Fig. 2). The presence of Cu is discerned from the elemental maps that show the metal to be uniformly distributed over the SiO₂ particle. The textural properties of 10 wt% Cu/SiO₂-AE samples after calcination at 400–800 °C for 6–12 h are given in Table S1. The presence of CuO could be observed in the XRD pattern only after the sample was calcined at 700 °C for 6 h (Fig. S6). Clearly, the ammonia evaporation method provides an effective route to the synthesis of CuO particles with excellent resistance to agglomeration and particle growth.

3.2. Catalytic activity for dehydrogenation of ethanol

3.2.1. Effect of support and synthesis method

The role of the support was clarified by comparing 10 wt% Cu wet impregnated onto ZrO₂, La₂O₃, MgO, mesoporous MSU-9 (Si/Al 9) and SiO₂. In each case, the support itself had very low activity for ethanol dehydrogenation with less than 5% conversion at 260 °C (Fig. 3). However, after loading with Cu, ethanol conversion was confirmed for all samples. Although Cu/MSU-9 showed a high conversion of 78%, the selectivity to acetaldehyde was very low. The presence of Al in silica results in an acidic support that favors ethanol dehydration to ethene rather than the desired dehydrogenation reaction which takes place on weak acid or basic sites [77–79]. Hence, the best catalyst was Cu/SiO₂ with 78% conversion at 260 °C and ~98.5% selectivity to acetaldehyde. The other samples had either low activity or low selectivity.

Different methods were used to load copper (10 wt%) onto SiO₂. The highest activity was shown by Cu/SiO₂ synthesized by the ammonia evaporation (AE) method (Fig. 4). The catalyst maintained its high conversion at reaction temperatures up to 300 °C, showing its excellent thermal stability. The catalyst prepared by DP-AE also had good activity and stability but below 260 °C, its

Table 2
Textural properties of 600 °C-calcined Cu/SiO₂ samples.

Catalyst	Cu ^a (wt.%)	Cry. size ^b (nm)	Cu dispersion (%)	BET(m ² /g)	V _{pore} (cm ³ /g)
SiO ₂	–	–	–	406	2.30
0.5% Cu/SiO ₂ -AE	0.7	–	75.3	385	2.13
1.0% Cu/SiO ₂ -AE	1.1	–	58.1	366	1.94
2.5% Cu/SiO ₂ -AE	2.7	–	–	319	1.83
5.0% Cu/SiO ₂ -AE	5.5	–	–	308	1.78
10% Cu/SiO ₂ -AE	10.2	–	38.7	357	1.45
15% Cu/SiO ₂ -AE	15.1	–	–	358	0.80
20% Cu/SiO ₂ -AE	19.6	–	33.6	414	0.79
30% Cu/SiO ₂ -AE	31.6	–	–	383	0.44
40% Cu/SiO ₂ -AE	41.3	–	–	340	0.39
50% Cu/SiO ₂ -AE	50.8	13.8	–	232	0.35
10% Cu/SiO ₂ -WI	10.0	83.5	–	311	1.14
10% Cu/SiO ₂ -DP	10.5	22.8	–	276	0.74
10% Cu/SiO ₂ -DP-AE	10.3	–	–	305	0.71

^a Cu loading determined by ICP-AES.

^b Cu crystallite size calculated from XRD.

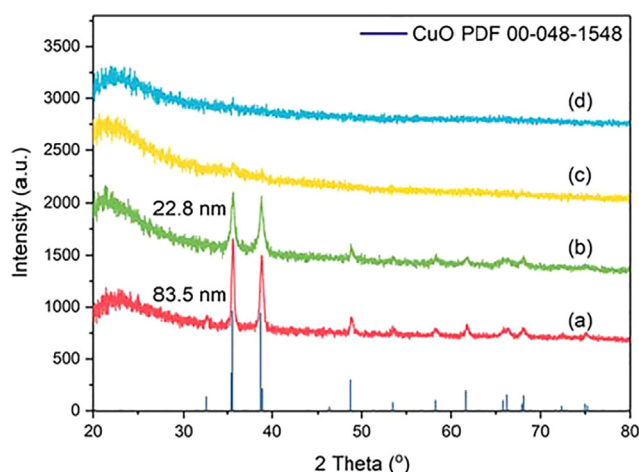


Fig. 1. X-ray diffractograms of Cu/SiO₂ synthesized by (a) WI (b) DP (c) DP-AE (d) AE.

conversion was slightly lower than that of Cu/SiO₂-AE. The sample prepared by deposition precipitation (DP) with Na₂CO₃ without the subsequent step of ammonia complexation and evaporation lost activity with time at 300 °C. The wet impregnated (WI) catalyst had the lowest activity with a rapid decline in activity above 240 °C. The poor stability may be due to sintering and/or carbon deposition, as there is no discernible difference in the basicity of the catalysts (Fig. S7). From these studies, Cu/SiO₂ prepared by ammonia evaporation was the best catalyst.

3.2.2. Effect of loading

Increasing the Cu loading from 0.5 to 50 wt% at constant W/F (catalyst weight/volume flow rate) led to increasingly higher conversions that levelled off above 20 wt% Cu (Fig. 5a). With the catalyst weight fixed, a higher loading offers more Cu sites for reaction although the intrinsic activity per site is lower. This can be seen from a comparison of the productivity, expressed as ethanol converted normalized to the Cu content ($g_{\text{ethanol}}/g_{\text{Cu}}\cdot\text{h}$), for the different loadings (Table S2). At 240 °C, the productivity fell from 93 to ~3 $g_{\text{ethanol}}/g_{\text{Cu}}\cdot\text{h}$ as the Cu loading increased from 0.5 to 50 wt %. This trend correlates with the decrease in Cu dispersion from 75.3% for 0.5% Cu/SiO₂-AE to 33.6% for 20 wt% Cu/SiO₂-AE, as determined by N₂O titration (Table 2). Small Cu particles contain a higher density of coordinatively unsaturated sites (*cus*) such as corners and kinks which are more reactive than steps or terrace sites [80]. However, the 0.5 and 1.0 wt% Cu/SiO₂-AE samples were not

thermally stable as the ethanol conversion at 280 °C decreased rapidly within a few hours (Fig. S8). No Cu peaks could be observed in the XRD patterns of the used catalysts (Fig. S9), ruling out sintering as the cause for deactivation. The absence of any observable sintering is remarkable, given the low Hüttig (134 °C) and Tamman temperatures (406 °C) of copper. At the Hüttig temperature, surface atoms at dislocations and defects start to diffuse over the surface, while at the Tamman temperature, bulk atoms become mobile. With minimal sintering, a likely cause for the loss in activity is carbon deposition.

Samples with Cu loadings ≥ 2.5 wt% showed no decrease in activity even when the reaction was carried out at 300 °C. As the catalytic activity depends on the availability of Cu sites at the surface, any carbon deposition will affect samples with low Cu loadings more severely than those with higher loadings [49]. For all samples, the selectivity to acetaldehyde was > 99% at low reaction temperatures and decreased slightly with temperature as some ethyl acetate was formed (Fig. 5b). From the temperature dependence, an apparent activation energy was calculated using the Arrhenius equation. The apparent activation energy was ~68 kJ/mol for 0.5% Cu/SiO₂-AE, but decreased to ~31 kJ/mol for ≥ 20 wt% Cu (Fig. S10). Previously reported activation energies for ethanol dehydrogenation on various catalysts are in the range of 48 to 59 kJ/mol [4,55,81,82]. The dependence of the activation energy on the metal loading suggests an implicit role of copper in the reaction mechanism. As discussed above, the dehydrogenation of alcohols is proposed to proceed via the initial cleavage of the hydroxyl bond to form the adsorbed alkoxide and hydrogen-bonded intermediate, followed by the removal of an α -H from the alkoxide to yield the carbonyl compound and H₂ [44,47–51]. Kinetic isotope studies have shown that the latter step is rate-determining [29,80]. Activation of ethanol to ethoxide can occur at the Cu surface or at the SiO₂ surface. As the α -H abstraction is catalysed by the metal, ethoxides that adsorbed at the interface between SiO₂ and Cu can undergo C-H bond cleavage to form H₂ and acetaldehyde. Regardless of whether ethanol is adsorbed on the metal cluster or on the support, a higher Cu loading offers more sites and more interfacial boundary for α -H abstraction. Hence, 10 wt% Cu loading was used for further investigations as it gives good productivity and robustness against deactivation without excessive metal content.

3.2.3. Effect of calcination temperature

To find the optimal calcination conditions, 10 wt% Cu/SiO₂-AE was treated at temperatures between 300 and 800 °C for 6 to 12 h. The conversion and acetaldehyde selectivity remained largely unchanged despite the widely differing calcination conditions

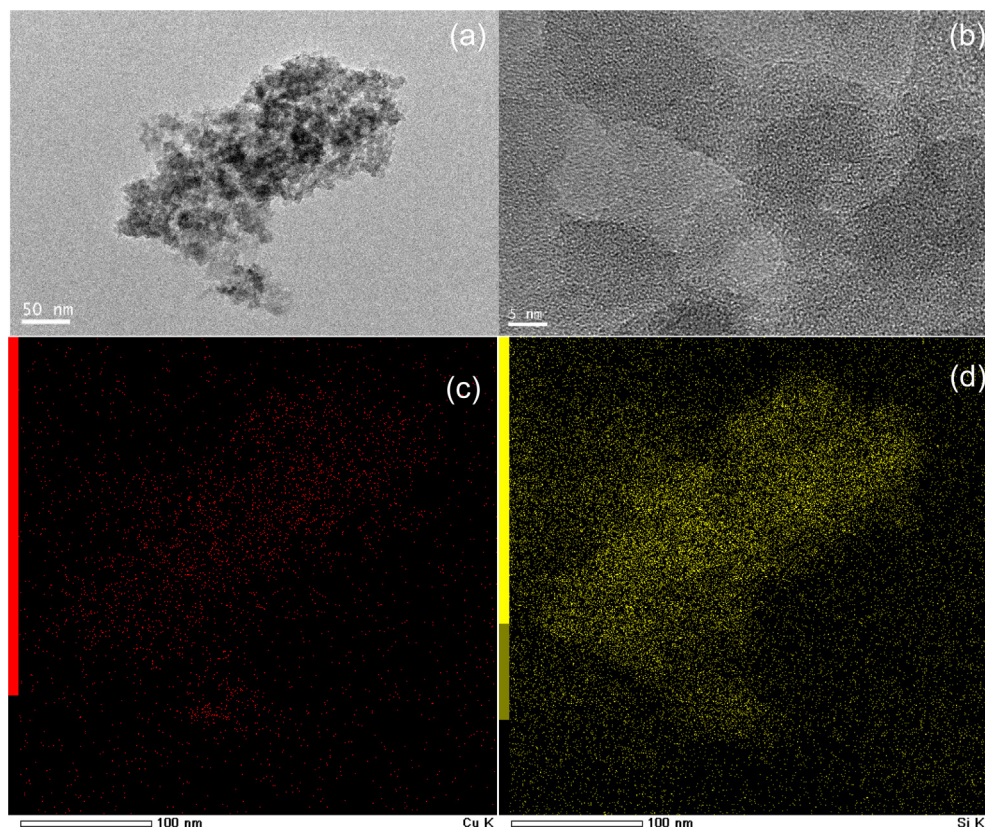


Fig. 2. (a, b) Transmission electron micrographs and (c, d) elemental maps of Cu and Si of 10 wt% Cu/SiO₂-AE.

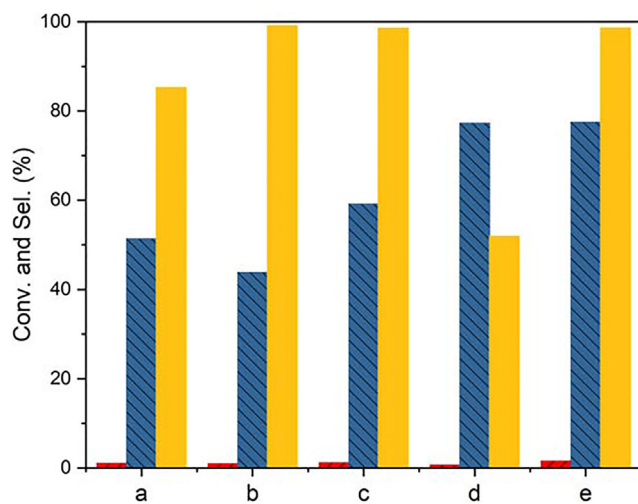


Fig. 3. Dehydrogenation of ethanol over Cu supported on (a) ZrO₂ (b) La₂O₃ (c) MgO (d) MSU-9 and (e) SiO₂. Ethanol conversion for (■) support only, (■) 10 wt% Cu loading, and (■) acetaldehyde selectivity. Reaction condition: 0.1 g catalyst, 1.5 g glass beads, 30 ml/min He, 200 μ l/h ethanol (4.37 vol% ethanol), 260 $^{\circ}$ C.

(Fig. S11). As the top priority is a long catalyst lifetime, the calcination conditions were chosen to minimize any changes in the phase of support or dispersion of the metal during prolonged time-on-stream. Hence, a high calcination temperature of 600 $^{\circ}$ C and a long heating duration of 12 h was chosen to ensure textural stability of the catalyst. With a lower calcination temperature or shorter duration, the crystalline phase of the support and the metal particle size may change with time.

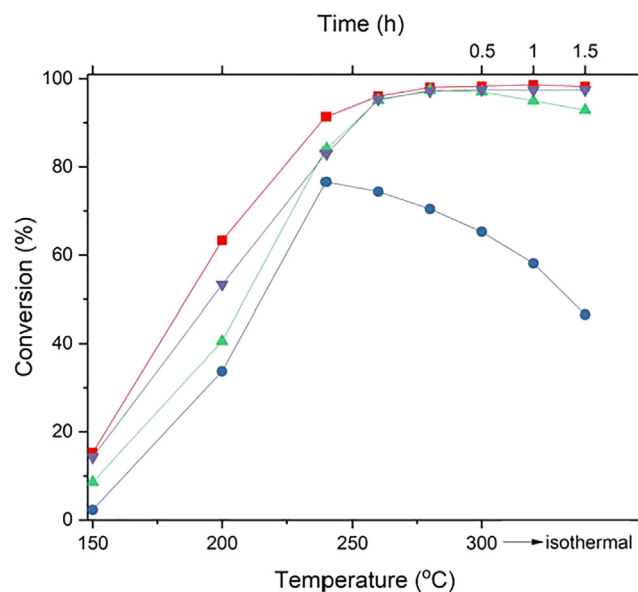


Fig. 4. Activity for ethanol dehydrogenation over 10 wt% Cu/SiO₂ catalysts synthesized by (■) AE (▼) DP-AE (▲) DP and (●) WI. Reaction condition: 0.1 g catalyst, 1.5 g glass beads, 30 ml/min He, 200 μ l/h ethanol (4.37 vol% ethanol).

Despite calcination at 600 $^{\circ}$ C for 12 h, no CuO particles could be detected by XRD or TEM for 0.5 – 30 wt% Cu/SiO₂-AE samples (Fig. S4). However, after reduction at 300 $^{\circ}$ C, very small particles of metallic copper of 1.5 to 2.9 nm were observed by TEM (Figs. 6a and S12) but not by XRD as they are below the detection limit (Fig. S13). For the 10 wt% Cu sample, the average size (based

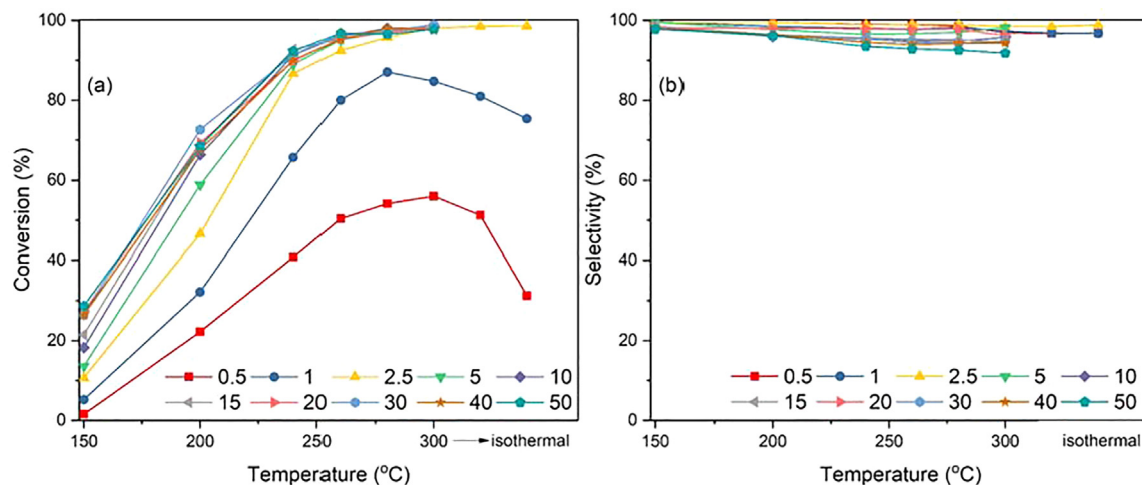


Fig. 5. (a) Effect of temperature on ethanol conversion and (b) selectivity to acetaldehyde over Cu/SiO₂-AE catalysts with different Cu loading (0.5 – 50 wt%). Reaction condition: 0.1 g catalyst, 1.5 g glass beads, 30 ml/min He, 200 μ l/h ethanol (gas phase composition: 4.37 vol% ethanol).

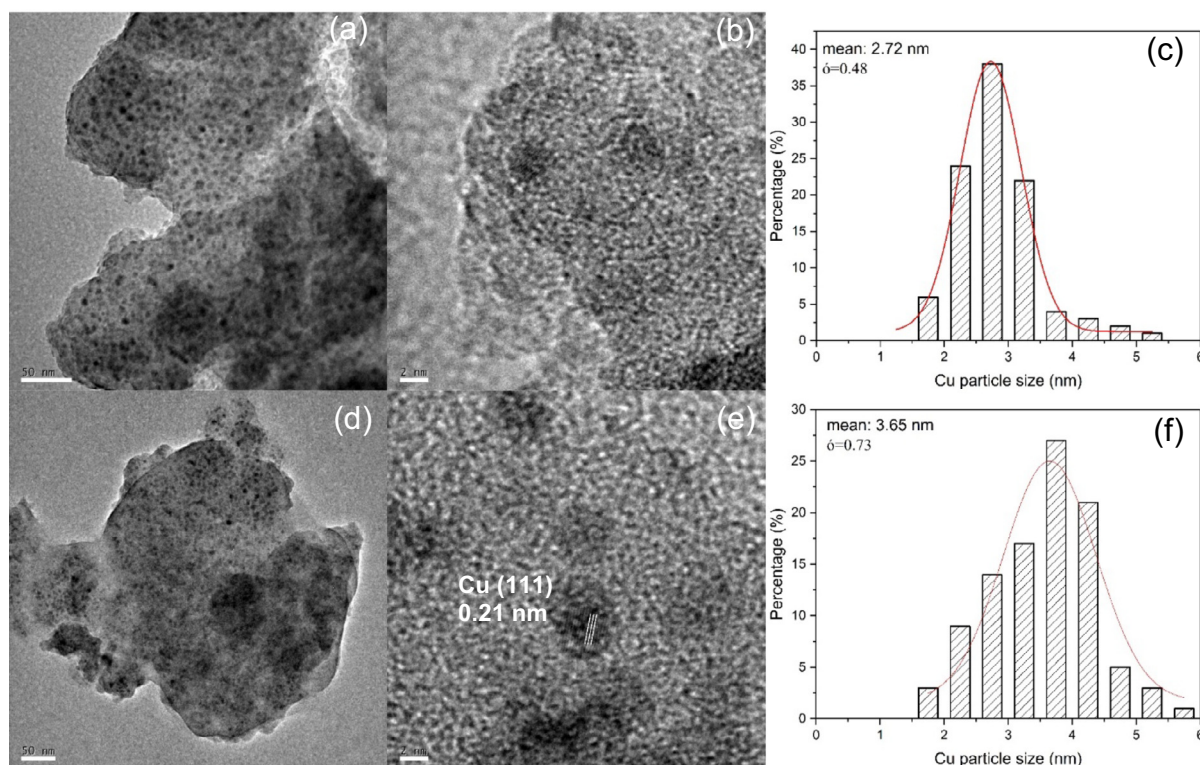


Fig. 6. Transmission electron micrographs and Cu particle size distribution of (a-c) reduced 10 wt% Cu/SiO₂ and (d-f) after 500 h time-on-stream.

on 100 particles) increased only slightly from ~ 2.7 nm to 3.6 nm after 500 h on-stream, showing the excellent stability of the Cu particles (Fig. 6d-f).

3.2.4. Effect of pretreatment

A 10 wt% Cu/SiO₂-AE sample was subjected to a flow of 10 vol% H₂/He at 300 °C for 10 – 60 min. Irrespective of the length of reduction, there was no obvious difference for either activity or selectivity. Hence, a short pre-reduction of 10 min was sufficient to form an active catalyst. Even without any pre-treatment, the activity of the as-calcined catalyst rose from 60% at 10 min to a steady level of 97% after 0.5 h of time-on-stream (Fig. S14). According to Campisano et al., ethanol dehydrogenation does not occur on Cu²⁺ [54].

Being weakly acidic, the pure SiO₂ support has only a very low activity for ethanol dehydrogenation. Nevertheless, the small amount of evolved hydrogen is able to reduce CuO to metallic Cu and initiates an autocatalytic reaction until the CuO is fully reduced. For complete reduction of the copper in 0.1 g of catalyst with 10 wt% Cu, 0.157 mmol of hydrogen is required. This amount can be generated within 5.5 min (assuming 50% ethanol conversion at 200 μ l/h).

The H₂-TPR profiles of 10 wt% Cu/SiO₂ show that irrespective of the synthesis method, the copper oxide species in the catalysts could be reduced at temperatures as low as 220 °C (Fig. S15). The TPR profiles showing only a single peak were similar to those reported for other supported Cu catalysts [56,67,83]. The low

reduction temperature has been ascribed to well-dispersed CuO nanoparticles [84]. No further reduction occurred at higher temperatures, indicating the absence of bulk CuO which is reduced in the range of 255–300 °C [26]. The H₂ consumption of 1.58 mmol/g for 10 wt% Cu/SiO₂-AE is in excellent agreement with the amount calculated (1.57 mmol/g) for the complete reduction of the CuO to metallic Cu. Parameters such as particle size, support and preparation method can influence the reducibility of copper. Dandekar et al found that smaller Cu crystallites supported on different types of carbon were easier to reduce than bigger ones [85]. In contrast, EXAFS measurements by Bueno's group reported that under hydrogen or water gas shift reaction conditions at 300 °C, the Cu-O coordination number of Cu/Al₂O₃ catalysts decreased with increasing particle size, indicating more facile reduction of the bigger particles [86]. Knapp et al. observed a similar trend from XANES results for Cu supported on alumina with varying surface areas; as the Cu particle size increased from 0.7 to 2.1 nm, the degree of reduction rose from 0.66 to 0.95 [87]. In addition, the Cu catalysts underwent further reduction during the water-gas shift reaction. Similarly, we noted that the active reduced catalyst can be generated *in-situ* during reaction. Hence, a separate pre-reduction of the catalyst is not required which is advantageous for industrial operation.

XPS measurements were carried out to gain more insight into the chemical state of the surface copper. For the calcined 10 wt% Cu/SiO₂-AE sample, the Cu 2p_{3/2} and 2p_{1/2} at binding energies are 935.5 and 955.5 eV, respectively, and the accompanying shakeup satellites show that copper was present as Cu²⁺ (Fig. 7a). However, after reduction at 300 °C, the Cu 2p_{3/2} was shifted to 932.7 eV which falls within that expected for Cu⁰ and Cu⁺. Given the broadness of the peak, the assignment of the oxidation state is ambiguous [58,65]. Hence, the modified Auger parameter, α' , defined as the sum of the kinetic energy of the Cu LMM Auger electron and the binding energy of the Cu 2p_{3/2} photoelectron, was calculated [88]. The α' for Cu⁰ is 1851.0 eV and for Cu⁺ is 1849.0 eV, which allows to distinguish between them [67]. The α' value for the reduced 10 wt% Cu/SiO₂-AE is 1850.7 eV which agrees with that for metallic Cu. The symmetrical shape of the Cu LMM peak with maximum at 918 eV further supports the existence of only one

rather than two copper species. For samples with lower Cu loadings of 1, 2.5, and 5 wt%, the XPS results also showed that after reduction at 300 °C, Cu⁰ was formed (Fig. S16). Similar findings have been reported by others for reduced Cu/SiO₂ catalysts prepared by deposition precipitation or ammonia evaporation method [71,83,84,89]. However, there are other studies that showed that the Cu/SiO₂ samples were reduced to a mixture of Cu⁺ and Cu⁰ [67,84]. The existence of Cu⁺ was attributed to electron transfer from the Cu to SiO₂ as a result of a strong interaction. Examination of the used catalyst after 500 h on-stream showed copper in the metallic state (Fig. 7a). Hence, it can be inferred that the active state during reaction was Cu⁰ which was maintained by the high partial pressure of hydrogen produced in the dehydrogenation of ethanol. It should be pointed out that the XPS spectra of the samples showed some Cu²⁺ to be present, as indicated by the peak tailing at ~935.5 eV and the shakeup satellite peak. This is likely due to inadvertent exposure to air during sample transfer to the XPS spectrometer. In-situ XRD measurements by Ge's group have shown that highly dispersed metallic copper is easily oxidized to Cu₂O or CuO when exposed to air [83].

3.2.5. Effect of GHSV and ethanol concentration

The dehydrogenation activity of 10 wt% Cu/SiO₂-AE at 280 °C with different ethanol concentrations and GHSV was investigated. With 2.23 vol% ethanol in the feed stream, the conversion was ~98% for GHSV from 1450 to 29000 h⁻¹ (Fig. 8a). The good performance can be attributed to the low concentration of ethanol and high capacity of the catalyst. When the ethanol concentration was increased to 4.37 vol%, the conversion was ~97.7% at a GHSV of 8895 h⁻¹ but it decreased to 93.6% for a higher GHSV of 29650 h⁻¹. At an even higher ethanol concentration of 6.42 vol%, the conversion was still ~93% at a GHSV of 30300 h⁻¹.

The contact time of the reactant, which is inversely proportional to the GHSV, is a more intuitive parameter to follow reaction details. As shown in Fig. 8(b), the conversion fell with higher ethanol concentrations but increased with longer contact time between the ethanol vapor and the catalyst bed. However, the acetaldehyde selectivity decreased with longer contact times due to an increasing probability for yield-reducing consecutive reactions. Neverthe-

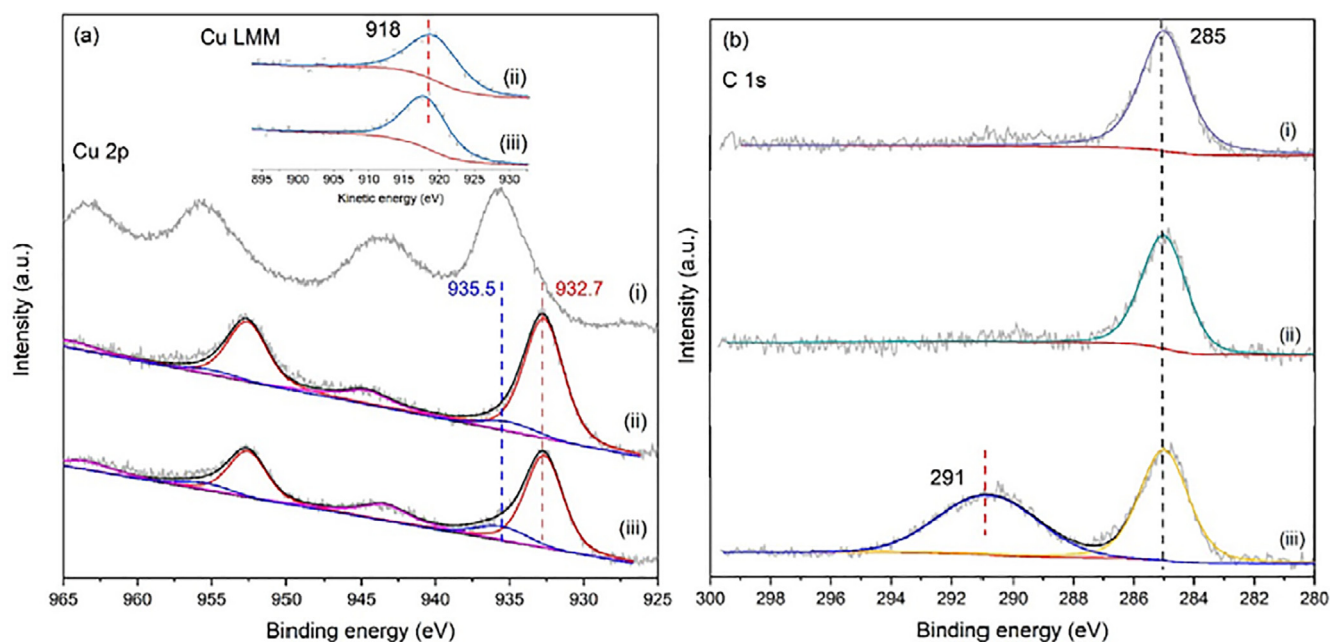


Fig. 7. XPS profiles of (a) Cu 2p and (b) C 1s of 10 wt% Cu/SiO₂-AE after (i) calcination at 600 °C (ii) reduction at 300 °C and (iii) reaction at 280 °C for 500 h on stream.

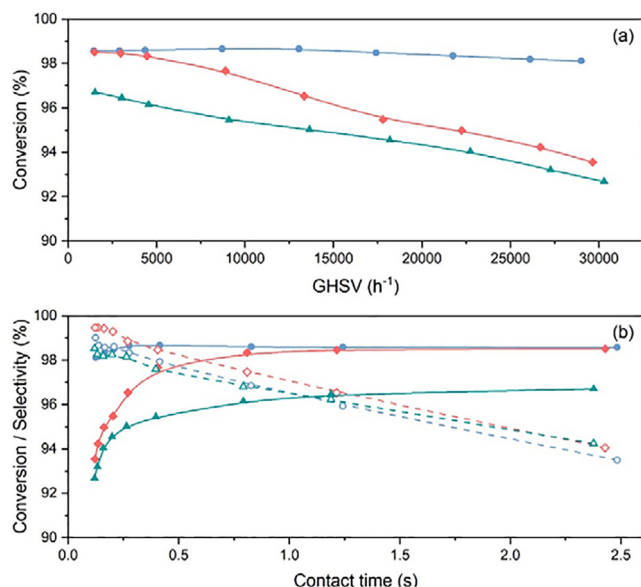


Fig. 8. Conversion (—) and acetaldehyde selectivity (---) of 10 wt% Cu/SiO₂-AE as a function of (a) GHSV and (b) contact time for (●) 2.24, (◆) 4.37 and (▲) 6.42 vol% ethanol. Reaction condition: 0.1 g Cu catalyst, 1.5 g glass beads, 5–100 ml/min He, 16.7–1000 μl/h ethanol, 280 °C.

less, the selectivity was still higher than 93% across all ethanol concentrations, even with contact times up to around 2.5 s.

3.2.6. Long-term stability of Cu/SiO₂ catalysts

The 10 and 20 wt% Cu/SiO₂ catalysts synthesized by the AE method exhibited excellent stability (Fig. 9). The ethanol conversion was > 90% with the selectivity to acetaldehyde and hydrogen at ~99% throughout the 500 h time-on-stream (Fig. S17). In contrast, the conversion over 10 wt% Cu/SiO₂-DP prepared by deposition–precipitation with Na₂CO₃ dropped to 56% within only 14 h (Fig. S18a). For 10 wt% Cu/SiO₂-DP-AE where the CuCO₃ precipitate was complexed with ammonia and re-precipitated, the conversion was maintained above 90% with 99% acetaldehyde selectivity for at least 100 h time-on-stream (Fig. S18b). It is evident that the ammonia-evaporation method is central for good catalyst stability. From previous studies, the presence of ammonia during the synthesis has been proposed to (i) lead to the formation of a copper-ammonia complex, [Cu(NH₃)₄]²⁺ and (ii) at the alkaline

pH, some silica is hydrolyzed to silicic acid, Si(OH)₄ [84]. Upon ammonia evaporation, [Cu(NH₃)₄]²⁺ dissociates and hydrolyzes to form Cu(OH)₂(H₂O)₄. This then undergoes condensation with the silicic acid to form copper phyllosilicate, Cu₂Si₂O₅(OH)₂, which is a key factor in the formation of nanosized metal particles with high thermal stability [73]. On heating to temperatures above 450 °C, copper phyllosilicate decomposes to CuO and SiO₂. Due to the strong interaction between the isolated copper ions and silica, the metal particles display remarkable resistance against aggregation even when heated to high temperatures.

Indeed, examination of our used AE catalysts revealed that no significant growth of the Cu particles had taken place despite the extended time-on-stream. A very broad peak at 2θ ~ 42° was observed by XRD signifying very small Cu crystallites (Fig. 9 inset). From TEM, the average size of the Cu particles was ~ 3.6 nm, only slightly bigger than at the start of the reaction (Fig. 6). Based on these observations, it is unlikely that sintering is the cause for any decline in the catalytic activity of the Cu/SiO₂-AE samples over time.

The XPS C1s spectrum of the used 10 wt% Cu/SiO₂-AE shows two peaks (Fig. 7b). The peak at a binding energy of ~ 285 eV is due to adventitious carbon contamination while the second peak at ~ 291 eV can be assigned to carbon from oxygenated coke formation during reaction [90]. Formation of polymeric acetaldehyde could be a plausible early step in coke lay-down [91]. To determine the location of the coke, TEM EDS was performed. From the elemental maps, the coke appeared to be distributed on the Cu particles as well as the SiO₂ support (Fig. S19). The amount of coke formed during reaction was quantified by TGA-MS in an oxidizing atmosphere. Three steps were observed in the weight loss profile: the first below 200 °C is accompanied by evolution of H₂O; in the second step between 200 and 600 °C, CO₂ and H₂O were detected in the effluent, indicating the presence of coke at the surface of the sample, and the third step around 800 °C, H₂O was evolved due to condensation of surface silanol groups (Fig. S20). For 0.5–2.5 wt% Cu/SiO₂, the second weight loss occurred up to 600 °C whereas it was already completed below 550 °C for 10 and 20 wt% Cu/SiO₂. The different weight loss profiles suggest variations in the nature of the carbon deposits on the samples. Depending on the loading, the Cu particles have different dispersion and size which would influence the activity and pathway for coke formation. From the weight loss over this temperature range, the amount of coke (CH_x) formed can be quantified (Table 3). This works out to 0.025 and 0.015 g CH_x/mmol Cu for the 500 h-used 10 and

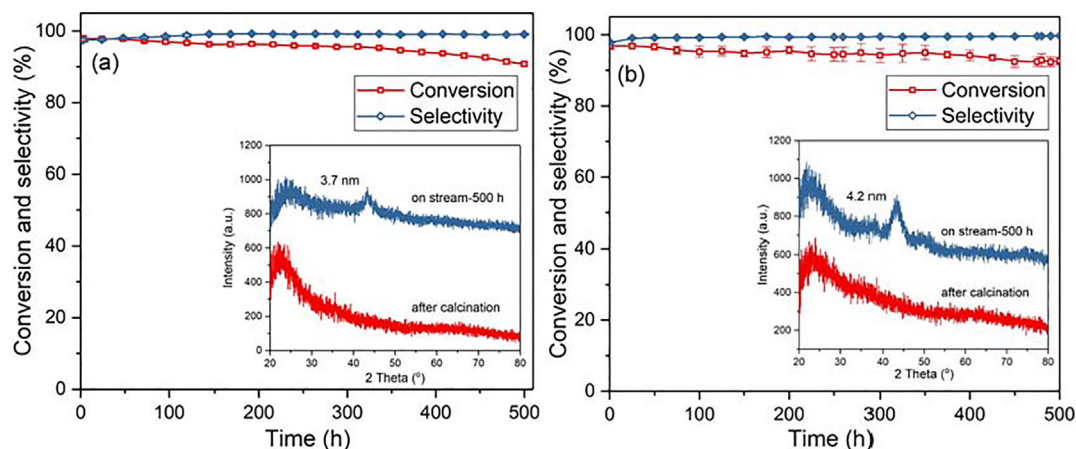


Fig. 9. Long term stability of (a) 10 and (b) 20 wt% Cu catalysts synthesized by AE method. Inset: XRD of samples after calcination and after 500 h time-on-stream reaction. Reaction condition: 0.1 g Cu catalyst, 1.5 g glass beads, 30 ml/min He, 200 μl/h ethanol (4.37 vol% ethanol), GHSV 8895 h⁻¹, 280 °C.

Table 3Weight loss from 200 to 600 °C for Cu/SiO₂-AE samples after reaction.^a

x (wt. % Cu)	Wt. loss (%)	CH ₃ /Cu (g/mmol)
0.5	1.73	0.22
1.0	2.05	0.13
2.5	2.34	0.06
10	3.89	0.025
20	4.70	0.015

^a Reaction condition: 0.1 g Cu catalyst, 1.5 g glass beads, 30 ml/min He, 200 μl/h ethanol (4.37 vol% ethanol), GHSV 8895 h⁻¹. For 0.5, 1 and 2.5 wt% Cu/SiO₂-AE, reaction from 150 to 280 °C and holding at 280 °C for 1.5 h. For 10 and 20 wt% Cu/SiO₂, reaction was carried out for 500 h on stream at 280 °C.

20 wt% Cu/SiO₂-AE, respectively. In comparison, much higher values (0.06–0.22 g CH₃/mmol Cu) were found for 0.5–2.5 wt% Cu/SiO₂ after only a few hours on-stream. Thus, it can be concluded that coke formation is especially pronounced at low metal loadings.

For a feed concentration of 4.37 vol% ethanol, there are 0.378 μmol of ethanol in the catalyst bed volume of 0.39 cm³ (Supplementary Material). Assuming that at least three Cu sites are required for the dissociative adsorption of ethanol to ethoxide with subsequent cleavage of the α-C-H, we calculate that a minimum of 1.14 μmol surface Cu atoms are needed. With 0.5 wt% loading on 0.1 g of catalyst, the total Cu corresponds to 7.87 μmol of which 5.93 μmol are at the surface (based on 75.3% dispersion determined by N₂O titration). Hence, the sites required for reaction constitute at least 19.3% of surface Cu atoms or 14.5% of the total Cu. In comparison, for a 10 wt% Cu/SiO₂ catalyst, the surface and total Cu atoms are 60.9 and 157 μmol, respectively. As the number of available Cu sites greatly exceeds the site requirement, the impact of coking on the catalytic activity of samples with higher Cu loading will be minimal. Furthermore, the coke can be easily removed by oxidation. Fig. 10 shows that the deactivation was reversed by heating the used 0.5 wt% Cu/SiO₂-AE in a flow of 30 vol% O₂ at 500 °C for 30 min. This oxidative treatment successfully regenerated the catalyst to full activity even after 3 cycles. The results of our study show that judicious choice of synthesis methodology and metal loading minimizes metal sintering and coking.

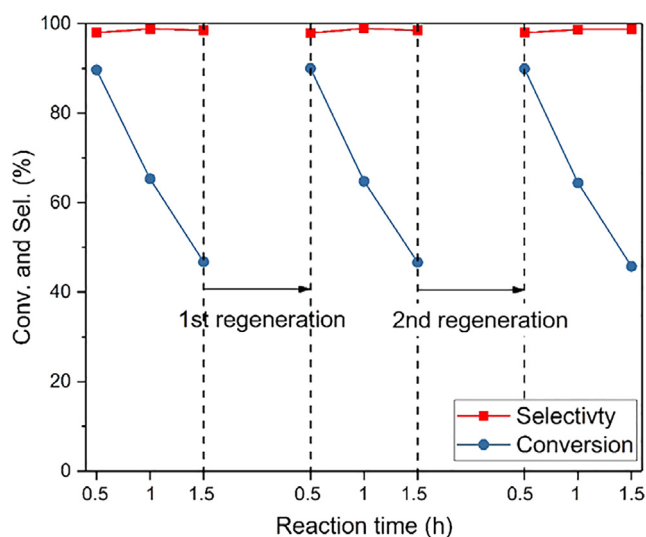


Fig. 10. Reuse of 0.5 wt% Cu/SiO₂-AE catalyst for 3 runs. Regeneration condition: oxidation at 500 °C for 30 min under 30 vol% O₂ followed by reduction at 300 °C for 10 min using 10 vol% H₂/He. Reaction condition: 0.1 g Cu catalyst, 1.5 g glass beads, 30 ml/min He, 200 μl/h ethanol (4.37 vol% ethanol), GHSV 8895 h⁻¹, 280 °C.

4. Conclusions

Three synthesis methods - ammonia evaporation, wet impregnation and deposition-precipitation - were evaluated for the synthesis of Cu/SiO₂ catalysts. The ammonia evaporation method resulted in highly active Cu/SiO₂ catalysts for ethanol dehydrogenation. The copper particles in these samples had a very small crystallite size and high dispersion even after calcination at temperatures up to 600 °C. The remarkable resistance to sintering was attributed to the formation of a copper phyllosilicate phase in the catalyst precursor which was identified by XRD and FTIR. Catalysts with 10 wt% Cu and higher maintained their excellent activity for ethanol dehydrogenation over an extended time-on-stream in contrast to those with 0.5–5 wt% Cu which deactivated within hours. The main cause of catalyst deactivation was found to be coke formation. The coke to Cu ratio in the used catalysts varied inversely with the Cu loading. Over a 10 wt% Cu/SiO₂-AE catalyst, high yields of acetaldehyde could be maintained for at least 500 h at 280 °C using a reaction mixture of 4.37 vol% ethanol at a GHSV of 8895 h⁻¹ (34930 ml g⁻¹h⁻¹). This performance exceeds that of previously reported catalysts and demonstrates impressively that an active and robust Cu/SiO₂ catalyst, resistant to deactivation, can be synthesized for the non-oxidative dehydrogenation of ethanol to acetaldehyde and hydrogen.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jcat.2020.05.018>.

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